

# Online Appendix for “Information Design for Differential Privacy”

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## S.1 Oblivious Mechanisms Without Anonymous Respondents

Here, we provide a counterexample showing that Theorem 2 does not hold when respondents are not anonymous.

Suppose that the database consists of the results of COVID-19 tests among three members of a university’s economics department, and that the decision maker is another faculty member who is interested in knowing the department’s positivity rate so that he can decide what kind of precautions to take. Thus, the data is binary (where 1 indicates a positive test and 0 indicates a negative test) and the population statistic  $\omega$  is a count.

Two of the tested faculty — respondents 2 and 3 — are married to one another, and so their test results are highly correlated. Specifically, conditional on the first two faculty members’ results  $(\theta_1, \theta_2)$ , the probability that  $\theta_3 = \theta_2$  is  $1 - \delta$  for some small  $\delta > 0$ ; i.e., we have

$$\begin{aligned} \beta_0((0,0,0)) = \beta_0((1,1,1)) = (1 - \delta)/3 \quad \text{and} \quad \beta_0((0,1,1)) = \beta_0((1,0,0)) = (1 - \delta)/6, \\ \text{but } \beta_0((0,0,1)) = \beta_0((1,1,0)) = \delta/3 \quad \text{and} \quad \beta_0((0,1,0)) = \beta_0((1,0,1)) = \delta/6. \end{aligned}$$

For small enough  $\delta$ , a non-oblivious mechanism can induce posteriors about the population statistic that are inaccessible to  $\epsilon$ -differentially private oblivious mechanisms without changing the level of privacy loss. In particular, in the limit  $\delta \rightarrow 0$ , a population statistic of  $\omega = 1$  *always* corresponds to the database  $(1,0,0)$ , while a population statistic of  $\omega = 2$  *always* corresponds to the database  $(0,1,1)$ . But these databases differ in *all three* entries, so differential privacy only indirectly restricts the amount that the posterior probability of one can differ from the posterior probability of the other. This allows the designer to provide much more information about whether the population statistic  $\omega$  is 1 rather than 2.

Specifically, writing each  $\beta \in \Delta(\{0,1\}^3)$  as the vector

$$\left[ \beta((0,0,0)) \quad \beta((1,0,0)) \quad \beta((0,1,0)) \quad \beta((0,0,1)) \quad \beta((1,1,0)) \quad \beta((1,0,1)) \quad \beta((0,1,1)) \quad \beta((1,1,1)) \right]',$$

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consider the posterior

$$\hat{\beta} = \frac{\phi \circ \beta_0}{\phi \cdot \beta_0}, \text{ where } \phi = \begin{bmatrix} e^{-2\epsilon} & e^{-3\epsilon} & e^{-\epsilon} & e^{-\epsilon} & e^{-2\epsilon} & e^{-2\epsilon} & 1 & e^{-\epsilon} \end{bmatrix}',$$

where  $\circ$  denotes the elementwise (Hadamard) product. This posterior is  $\epsilon$ -differentially private:  $\hat{\beta} \in K(\epsilon, \beta_0)$ . It achieves the upper privacy bound  $\frac{\hat{\beta}((0,0,0))/\beta_0((0,0,0))}{\hat{\beta}((1,0,0))/\beta_0((1,0,0))} = e^\epsilon$  between  $(0,0,0)$  and  $(1,0,0)$ , and the lower privacy bound  $\frac{\hat{\beta}((1,1,1))/\beta_0((1,1,1))}{\hat{\beta}((0,1,1))/\beta_0((0,1,1))} = e^{-\epsilon}$  between  $(0,1,1)$  and  $(1,1,1)$ , while achieving the privacy bounds between other databases in a way that distinguishes between  $(1,0,0)$  and  $(0,1,1)$  as much as possible.<sup>1</sup> As  $\delta \rightarrow 0$ , its third through sixth entries vanish along with the corresponding entries of  $\beta_0$ . Hence, its projection  $P\hat{\beta}$  onto  $\Delta(\Omega)$  approaches

$$\hat{\mu} = \frac{\psi \circ \mu_0}{\psi \cdot \mu_0}, \text{ where } \psi = \begin{bmatrix} e^{-2\epsilon} & e^{-3\epsilon} & 1 & e^{-\epsilon} \end{bmatrix}'.$$

This posterior about the population statistic is outside of  $K_\Omega(\epsilon, \mu_0)$ , and so cannot be induced with an oblivious mechanism: it exceeds the upper privacy bound  $\frac{\mu(2)/\mu_0(2)}{\mu(1)/\mu_0(1)} \leq e^\epsilon$  between states  $\omega = 1$  and  $\omega = 2$ . Consequently, when  $\delta$  is small enough, and  $\theta_2$  and  $\theta_3$  are very highly correlated, oblivious mechanisms are not always optimal.<sup>2</sup>

## S.2 Differential Privacy and Learning

Here, we offer an interpretation of differential privacy as a bound on Bayesian updating. Proposition S.1 shows that differential privacy is equivalent to a bound on the amount that learning the mechanism's output can cause an observer who believes respondents' types are independent to update their beliefs about a specific respondent's type  $\theta_n$ . In particular, differential privacy limits the proportional change in the odds of any one type relative to another. The argument mirrors that of Theorem 6.1 in Kifer and Machanavajjhala (2014). However, Proposition S.1 differs in that it bounds the change in the odds that the respondent has one type instead of another, rather than the change in the odds that it has a certain type instead of being absent from the data altogether.

**Proposition S.1** (Interpretations of Differential Privacy). *The following are equivalent:*

- i.  $(S, \pi)$  is  $\epsilon$ -differentially private.
- ii. For every respondent  $n \in \{1, \dots, N\}$  and every pair of distinct types  $t, t' \in \{0, \dots, T\}$ , if an agent's prior  $\hat{\beta}_0 \in \Delta(\Theta)$  is a product distribution that places positive probability on both  $\theta_n = t$  and  $\theta_n = t'$ ,

<sup>1</sup>Specifically, it achieves the upper privacy bound between  $(1,0,0)$  and both  $(1,1,0)$  and  $(1,0,1)$ , between  $(0,0,0)$  and both  $(0,1,0)$  and  $(0,0,1)$ , between  $(0,1,0)$  and  $(0,1,1)$ , between  $(0,0,1)$  and  $(0,1,1)$ , between  $(1,1,0)$  and  $(1,1,1)$ , and between  $(1,0,1)$  and  $(1,1,1)$ ; and the lower privacy bound between  $(0,1,0)$  and  $(1,1,0)$  and between  $(0,0,1)$  and  $(1,0,1)$ .

<sup>2</sup>Consider, for instance, a decision maker who takes action  $z$  when his belief about the state is in  $K_\Omega(\epsilon, \mu_0)$ , but takes a different action,  $x$ , when his belief about the population statistic is in some neighborhood of  $\hat{\mu}$ . (To see how this might occur, suppose that distinguishing between  $\omega = 1$  and  $\omega = 2$  is important for the decision maker's choice, but distinguishing between the other values of  $\omega$  is not, e.g., because action  $x$  gives a much worse payoff than action  $z$  when  $\omega = 1$ , a somewhat better payoff when  $\omega = 2$ , and the same payoff when  $\omega \in \{0, 3\}$ .) Then any oblivious  $\epsilon$ -differentially private mechanism does not offer any useful information to the decision maker, but a non-oblivious  $\epsilon$ -differentially private mechanism that induces  $\hat{\beta}$  does.

then after observing any output  $s$  from  $(S, \pi)$  with  $\pi(s|\theta) > 0$  for some  $\theta \in \text{supp } \hat{\beta}_0$ , the agent's posterior  $\hat{\beta}$  satisfies

$$\left| \log \left( \frac{\hat{\beta}(\{\theta : \theta_n = t\})}{\hat{\beta}(\{\theta : \theta_n = t'\})} \right) - \log \left( \frac{\hat{\beta}_0(\{\theta : \theta_n = t\})}{\hat{\beta}_0(\{\theta : \theta_n = t'\})} \right) \right| \leq \epsilon.$$

*Proof.* ((i)  $\Rightarrow$  (ii)): Suppose that  $(S, \pi)$  is differentially private and that an agent's prior  $\hat{\beta}_0$  is a product distribution, and fix  $n$  and  $\hat{\beta}_0$  as in (ii). Every  $s \in S$  has positive probability at some database, so differential privacy along a path of adjacent databases implies  $\pi(s|\theta) > 0$  for every  $\theta \in \Theta$ . Then we can write

$$\hat{\beta}(\{\theta : \theta_n = t\}) = \frac{\sum_{\theta: \theta_n = t} \pi(s|\theta) \hat{\beta}_0(\theta)}{\sum_{\theta \in \Theta} \pi(s|\theta) \hat{\beta}_0(\theta)} = \frac{\sum_{\theta: \theta_n = t} \pi(s|\theta) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})}{\sum_{\theta \in \Theta} \pi(s|\theta) \hat{\beta}_0(\theta)}$$

Hence, for any  $t, t' \in \{0, \dots, T\}$  with  $\hat{\beta}_0(\{\theta : \theta_n = t\}) > 0$  and  $\hat{\beta}_0(\{\theta : \theta_n = t'\}) > 0$ , we have

$$\begin{aligned} \left| \log \left( \frac{\hat{\beta}(\{\theta : \theta_n = t\})}{\hat{\beta}(\{\theta : \theta_n = t'\})} \right) - \log \left( \frac{\hat{\beta}_0(\{\theta : \theta_n = t\})}{\hat{\beta}_0(\{\theta : \theta_n = t'\})} \right) \right| &= \left| \log \left( \frac{\sum_{\theta: \theta_n = t} \pi(s|\theta) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})}{\sum_{\theta: \theta_n = t'} \pi(s|\theta) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})} \right) \right| \\ &= \left| \log \left( \frac{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t, \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})}{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t', \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})} \right) \right|. \end{aligned}$$

Now we have

$$\begin{aligned} &\left( \min_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\} \right) \left( \sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t', \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\}) \right) \\ &\leq \sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t, \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\}) \\ &\leq \left( \max_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\} \right) \left( \sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t', \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\}) \right), \end{aligned}$$

and so

$$\begin{aligned} \min_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\} &\leq \frac{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t, \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})}{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t', \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})} \leq \max_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\}; \\ \Rightarrow \left| \log \left( \frac{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t, \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})}{\sum_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \pi(s|(t', \theta_{-n})) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t'\}) \hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\})} \right) \right| \\ &\leq \max \left\{ \left| \log \left( \min_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\} \right) \right|, \left| \log \left( \max_{\theta_{-n} \in \{0, \dots, T\}^{N-1}} \left\{ \frac{\pi(s|(t, \theta_{-n}))}{\pi(s|(t', \theta_{-n}))} \right\} \right) \right| \right\} \leq \epsilon, \end{aligned}$$

as desired.

((ii)  $\Rightarrow$  (i)): Fix  $s \in S$ , let  $n \in \{1, \dots, N\}$  and let  $\theta, \theta' \in \Theta$  be such that  $\theta_{-n} = \theta'_{-n}$  and  $\pi(s|\theta) + \pi(s|\theta') > 0$ . (Such a pair exists since  $\pi(s|\theta) > 0$  for some  $\theta \in \Theta$ .) Let  $\hat{\beta}_0$  be such that  $\hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_{-n} = \theta_{-n}\}) = 1$  and  $\hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) \in (0, 1)$  for each  $t \in \{\theta_n, \theta'_n\}$ . Then for each  $t \in \{\theta_n, \theta'_n\}$ ,

$\hat{\beta}_0(\{\hat{\theta} : \hat{\theta}_n = t\}) = \hat{\beta}_0(t, \theta_{-n})$  and  $\hat{\beta}(\{\hat{\theta} : \hat{\theta}_n = t\}) = \frac{\pi(s|(t, \theta_{-n}))\hat{\beta}_0(t, \theta_{-n})}{\sum_{k=0}^T \pi(s|(k, \theta_{-n}))\hat{\beta}_0(k, \theta_{-n})}$ . Hence

$$\epsilon \geq \left| \log \left( \frac{\hat{\beta}(\theta)}{\hat{\beta}(\theta')} \right) - \log \left( \frac{\hat{\beta}_0(\theta)}{\hat{\beta}_0(\theta')} \right) \right| = \left| \log \left( \frac{\pi(s|\theta)}{\pi(s|\theta')} \right) \right|. \quad (1)$$

As a consequence, we must have  $\pi(s|\theta) > 0$  and  $\pi(s|\theta') > 0$ . Since we can construct a chain of adjacent databases connecting any two databases, it follows that  $\pi(s|\hat{\theta}) > 0$  for each  $\hat{\theta} \in \Theta$ . Therefore, (1) holds for every pair of adjacent databases  $\theta, \theta' \in \Theta$ . (i) follows.  $\square$

### S.3 Discreteness and Oblivious Mechanisms

Here, we give a counterexample showing that the gap between  $K_\Omega(\epsilon, \mu_0)$  and  $PK(\epsilon, \beta_0)$  is not about the discrete set of values that each respondent's type  $\theta_i$  can take. In particular, it need not vanish as  $T \rightarrow \infty$ .

Since we are interested in this limit, suppose  $T \geq N$ , and for each such  $T$ , let  $\beta_0$  be uniform over the  $(T+1)^N$  databases, so that respondents' types are independent and uniform on  $\{0, \dots, T\}$ . Let  $d(\theta, \theta') \equiv |\{i | \theta_i \neq \theta'_i\}|$  be the Hamming distance, and fix  $\hat{\theta} = (1, \dots, 1)$ , so that  $\omega_{\hat{\theta}} = N$ . Following the proof of Theorem 1, define

$$\psi(\theta) \equiv \begin{cases} \min\{N, d(\theta, \hat{\theta})\}, & \omega_\theta \neq N; \\ \min\{N-1, d(\theta, \hat{\theta})\}, & \omega_\theta = N, \end{cases} \quad Z_T \equiv \sum_{t \in \Theta} \beta_0(t) e^{-\epsilon \psi(t)}, \quad \beta_{\hat{\theta}}(\theta) \equiv \frac{\beta_0(\theta) e^{-\epsilon \psi(\theta)}}{Z_T}.$$

For any adjacent databases  $\theta, \theta' \in \Theta$ , changing one entry changes  $d(\cdot, \hat{\theta})$  by at most one. Therefore,  $|\psi(\theta) - \psi(\theta')| \leq 1$ , and

$$\left| \log \left( \frac{\beta_{\hat{\theta}}(\theta)}{\beta_{\hat{\theta}}(\theta')} \right) - \log \left( \frac{\beta_0(\theta)}{\beta_0(\theta')} \right) \right| = \epsilon |\psi(\theta) - \psi(\theta')| \leq \epsilon,$$

so  $\beta_{\hat{\theta}} \in K(\epsilon, \beta_0)$ . Moreover, for each  $\omega \in \Omega$ ,

$$\frac{P\beta_{\hat{\theta}}(\omega)}{\mu_0(\omega)} = \frac{1}{Z_T} \frac{\sum_{\theta: \omega_\theta = \omega} \beta_0(\theta) e^{-\epsilon \psi(\theta)}}{\sum_{\theta: \omega_\theta = \omega} \beta_0(\theta)} = \frac{1}{Z_T} \mathbb{E}_{\beta_0} \left[ e^{-\epsilon \psi(\theta)} \mid \omega_\theta = \omega \right].$$

Thus, up to the common factor  $1/Z_T$ , the likelihood ratio of the projected posterior to  $\mu_0$  at  $\omega$  is the average of  $e^{-\epsilon \psi(\theta)}$  over the databases with population statistic  $\omega$ .

Now since  $N \leq T$ , a change in one respondent's data can change  $\omega$  from 0 to  $N$ , so if  $P\beta_{\hat{\theta}} \in K_\Omega(\epsilon, \mu_0)$ , it must satisfy a privacy bound between  $\omega = 0$  and  $\omega = N$ . Only the database  $(0, 0, \dots, 0)$  has population statistic 0, and  $\psi(0, 0, \dots, 0) = N$ , so we have  $P\beta_{\hat{\theta}}(0)/\mu_0(0) = e^{-\epsilon N}/Z_T$ . For each of the  $\binom{2N-1}{N-1}$  databases  $\theta$  with  $\omega_\theta = N$ , on the other hand,  $\psi(\theta) \leq N-1$ , and in particular,  $\psi(\hat{\theta}) = 0$ . Thus, since  $\beta_0$  is uniform, we have

$$\frac{P\beta_{\hat{\theta}}(N)/\mu_0(N)}{P\beta_{\hat{\theta}}(0)/\mu_0(0)} = \mathbb{E}_{\beta_0} \left[ e^{(N-\psi(\theta))\epsilon} \mid \omega_\theta = N \right] \geq \frac{\left( \binom{2N-1}{N-1} - 1 \right) e^\epsilon + e^{N\epsilon}}{\binom{2N-1}{N-1}} = e^\epsilon \left( 1 + \frac{e^{(N-1)\epsilon} - 1}{\binom{2N-1}{N-1}} \right),$$

and  $P\beta_{\hat{\theta}}$  violates the privacy bound between these  $\omega = 0$  and  $\omega = N$  by at least  $\log(1 + (e^{(N-1)\epsilon} - 1)/(2_{N-1}^{N-1})) > 0$ . This bound does not depend on  $T$ , since the number of databases that share the population statistic  $N$  depends only on  $N$  once the type space is large enough to contain them all (i.e.,  $T \geq N$ ).

#### S.4 Permutation-Invariant Mechanisms

Here, we prove Proposition 4.

**Proof of Proposition 4 (Permutation-Invariant Mechanisms)** Let  $(S, \pi)$  be an  $\epsilon$ -DP mechanism, let  $G$  be the set of permutations of  $\{1, \dots, N\}$ , and for each  $g \in G$ , let  $g\theta \equiv (\theta_{g(1)}, \dots, \theta_{g(N)})$ . Define  $(S, \bar{\pi})$  by

$$\bar{\pi}(s|\theta) = \frac{1}{|G|} \sum_{g \in G} \pi(s|g\theta).$$

If  $\theta$  and  $\theta'$  differ in one entry, then  $g\theta$  and  $g\theta'$  also differ in one entry for every  $g \in G$ . Hence, for each  $s \in S$ , summing the differential privacy inequalities for these pairs over  $g \in G$  and dividing by  $|G|$  gives

$$\begin{aligned} e^{-\epsilon} \bar{\pi}(s|\theta') &= \frac{1}{|G|} \sum_{g \in G} e^{-\epsilon} \pi(s|g\theta') \leq \frac{1}{|G|} \sum_{g \in G} \pi(s|g\theta) = \bar{\pi}(s|\theta) \\ &\leq \frac{1}{|G|} \sum_{g \in G} e^{\epsilon} \pi(s|g\theta') = e^{\epsilon} \bar{\pi}(s|\theta'). \end{aligned}$$

Thus,  $(S, \bar{\pi})$  is  $\epsilon$ -DP. For each  $h \in G$ ,

$$\begin{aligned} \bar{\pi}(s|h\theta) &= \frac{1}{|G|} \sum_{g \in G} \pi(s|g(h\theta)) = \frac{1}{|G|} \sum_{g \in G} \pi(s|(h \circ g)\theta) \\ &= \frac{1}{|G|} \sum_{g' \in G} \pi(s|g'\theta) = \bar{\pi}(s|\theta), \end{aligned}$$

where the reindexing  $g' = h \circ g$  is valid because  $g \mapsto h \circ g$  is a bijection of  $G$ . Thus, the mechanism is permutation-invariant.

Finally, anonymity implies that  $\beta_0(g\theta) = \beta_0(\theta)$  for every  $g \in G$ , while  $\omega_{g\theta} = \omega_{\theta}$ . Thus, for each  $\omega \in \Omega$  and  $s \in S$ ,

$$\begin{aligned} \sum_{\theta: \omega_{\theta} = \omega} \beta_0(\theta) \bar{\pi}(s|\theta) &= \frac{1}{|G|} \sum_{g \in G} \sum_{\theta: \omega_{\theta} = \omega} \beta_0(g\theta) \pi(s|g\theta) \\ &= \sum_{\theta: \omega_{\theta} = \omega} \beta_0(\theta) \pi(s|\theta). \end{aligned}$$

Hence,  $(S, \pi)$  and  $(S, \bar{\pi})$  induce the same joint distribution of the population statistic and the output, and therefore the same distribution of posterior beliefs about the population statistic.  $\square$

## S.5 Zero-Concentrated Differential Privacy

Here, we consider *zero-concentrated differential privacy* (zCDP), an alternative privacy criterion introduced by Bun and Steinke (2016). Like differential privacy, zCDP limits the amount that changing a single respondent’s type can change the distribution of the mechanism’s realizations. The two criteria differ in how they treat unlikely realizations. Differential privacy (1) bounds the log likelihood ratio between adjacent databases at every output, no matter how improbable. zCDP instead bounds a family of averages of that likelihood ratio across outputs — the *Rényi divergences* between the mechanism’s output distributions at adjacent databases. This accommodates likelihood ratios between adjacent databases that would be inconsistent with DP, so long as the output in question is sufficiently unlikely. This difference allows zCDP to accommodate mechanisms whose likelihood ratios between adjacent databases are unbounded. For instance, adding mean-zero Gaussian noise with variance  $\sigma^2$  to the population statistic is not  $\epsilon$ -differentially private for any  $\epsilon$ , but satisfies  $\rho$ -zCDP for  $\rho = T^2 / (2\sigma^2)$ .<sup>3</sup>

zCDP is frequently used in practice. In particular, it is the formal privacy criterion adopted by the U.S. Census Bureau for the 2020 Decennial Census. The Bureau’s disclosure avoidance system produces the published tabulations from a collection of noisy measurements, formed by adding discrete Gaussian noise to counts computed from the confidential microdata, and accounts for the cumulative privacy loss from those measurements in terms of the zCDP parameter  $\rho$  (Abowd et al., 2022; Abowd and Hawes, 2023).

Below, we define zCDP in our model and characterize it as a restriction on the distribution of posterior beliefs induced by the publication mechanism. Unlike differential privacy, this restriction is not separable in those beliefs: that is, it does not reduce to a restriction on the *support* of the distribution of posteriors.

For distributions  $P, Q \in \Delta(S)$  and  $\alpha > 1$ , the *Rényi divergence of order  $\alpha$*  of  $P$  from  $Q$  is

$$D_\alpha(P\|Q) \equiv \frac{1}{\alpha - 1} \log \left( \sum_{s \in S} P(s)^\alpha Q(s)^{1-\alpha} \right), \quad (2)$$

with the conventions that  $0^{1-\alpha} = \infty$  and  $0 \cdot \infty = 0$ .

**Definition** ( $\rho$ -Zero-Concentrated Differential Privacy (Bun and Steinke, 2016)). A data publication mechanism  $(S, \pi)$  is  $\rho$ -zero-concentrated differentially private ( $\rho$ -zCDP) if for all databases  $\theta, \theta' \in \Theta$  that are adjacent (i.e., such that for some  $i \in \{1, \dots, N\}$ ,  $\theta_{-i} = \theta'_{-i}$ ) and all  $\alpha > 1$ , we have

$$D_\alpha(\pi(\cdot|\theta)\|\pi(\cdot|\theta')) \leq \rho\alpha. \quad (3)$$

Since (3) must hold for each ordered pair of adjacent databases, and Rényi divergence is not symmetric in its arguments, zCDP bounds both  $D_\alpha(\pi(\cdot|\theta)\|\pi(\cdot|\theta'))$  and  $D_\alpha(\pi(\cdot|\theta')\|\pi(\cdot|\theta))$ . Differential privacy can be written in the same terms:  $(S, \pi)$  satisfies (1) if and only if  $D_\infty(\pi(\cdot|\theta)\|\pi(\cdot|\theta')) \leq \epsilon$

<sup>3</sup>More generally, every  $\epsilon$ -differentially private mechanism is  $(\epsilon^2/2)$ -zCDP, and every  $\rho$ -zCDP mechanism is  $(\epsilon, \delta)$ -differentially private for each  $\delta \in (0, 1)$ , where  $\epsilon = \rho + 2\sqrt{\rho \log(1/\delta)}$  (Bun and Steinke, 2016). (A mechanism  $(S, \pi)$  is  $(\epsilon, \delta)$ -differentially private if  $\sum_{s \in E} \pi(s|\theta) \leq e^\epsilon \sum_{s \in E} \pi(s|\theta') + \delta$  for every  $E \subseteq S$  and each pair of adjacent databases  $\theta, \theta' \in \Theta$ .)

for each ordered pair of adjacent databases, where  $D_\infty(P\|Q) \equiv \sup_{s:P(s)>0} \log(P(s)/Q(s))$  is the limit of  $D_\alpha(P\|Q)$  as  $\alpha \rightarrow \infty$ .

Like differential privacy, zCDP can be reformulated in terms of the posterior beliefs that the mechanism induces. Let  $\beta_s \in \Delta(\Theta)$  denote the posterior belief induced by output  $s$ , and  $\nu \in \Delta(\Delta(\Theta))$  the distribution of posterior beliefs induced by the mechanism  $\pi$ ; without loss, assume that distinct outputs induce distinct posteriors.<sup>4</sup> Then Bayes' rule can be written as

$$\pi(s|\theta) = \nu(\beta_s) \frac{\beta_s(\theta)}{\beta_0(\theta)}, \quad (4)$$

and substituting (4) into (2) yields

$$D_\alpha(\pi(\cdot|\theta)\|\pi(\cdot|\theta')) = \frac{1}{\alpha-1} \log \left( E_\nu \left[ \left( \frac{\beta(\theta)}{\beta_0(\theta)} \right)^\alpha \left( \frac{\beta(\theta')}{\beta_0(\theta')} \right)^{1-\alpha} \right] \right) \quad (5)$$

for each pair of adjacent databases  $\theta, \theta' \in \Theta$  and each  $\alpha > 1$ .

The Rényi divergence thus depends on the publication mechanism only through the distribution of posteriors  $\nu \in \Delta(\Delta(\Theta))$  that the mechanism induces. This observation yields a counterpart of Lemma 4 for zCDP.

**Lemma S.1** (zCDP as a Constraint on Distributions of Posteriors). *The distribution  $\nu \in \Delta(\Delta(\Theta))$  of posterior beliefs can be induced with a  $\rho$ -zCDP mechanism if and only if  $\nu$  has countable support, is Bayes-plausible, and satisfies*

$$E_\nu \left[ \left( \frac{\beta(\theta)}{\beta_0(\theta)} \right)^\alpha \left( \frac{\beta(\theta')}{\beta_0(\theta')} \right)^{1-\alpha} \right] \leq e^{\rho\alpha(\alpha-1)} \text{ for each } \alpha > 1 \text{ and } \theta, \theta' \in \Theta \text{ with } \theta_{-n} = \theta'_{-n} \text{ for some } n. \quad (6)$$

*Proof.* Since the output space of a data publication mechanism is countable, every induced distribution of posteriors has countable support. Conversely, if  $\nu$  has countable support and is Bayes-plausible, define a mechanism with output space  $\text{supp } \nu$  by

$$\pi(\beta|\theta) = \nu(\beta) \frac{\beta(\theta)}{\beta_0(\theta)}.$$

Bayes-plausibility implies  $\sum_{\beta \in \text{supp } \nu} \pi(\beta|\theta) = 1$  for each  $\theta$ , and Bayes' rule implies that output  $\beta$  induces the posterior  $\beta$ . Thus,  $\nu$  can be induced with some data publication mechanism if and only if it has countable support and is Bayes-plausible. The claim then follows from (5) since  $\alpha - 1 > 0$  and log is increasing.  $\square$

Unlike differential privacy, which restricts only the *support* of the distribution of posteriors (Lemma 4), the constraints (6) weight each posterior by the probability with which it occurs. A posterior that shifts the log likelihood ratio between two adjacent databases by a large amount can thus be induced by a  $\rho$ -zCDP mechanism, provided that it is induced with sufficiently small probability. Consequently, zCDP does not have a direct counterpart of the set of  $\epsilon$ -DP posteriors.

<sup>4</sup>Merging outputs that induce the same posterior changes neither  $\nu$  nor the Rényi divergences (2).

Now consider oblivious mechanisms, which depend on the database only through the population statistic (Section 4). Proposition S.2 characterizes zCDP for these mechanisms; it is the counterpart of Proposition 2.

**Proposition S.2** (Zero-Concentrated Differential Privacy for Oblivious Mechanisms). *Suppose  $(S, \sigma)$  is an oblivious data publication mechanism that induces the distribution  $\tau \in \Delta(\Delta(\Omega))$  of posterior beliefs about the population statistic. Then the following are equivalent:*

- i.  $(S, \sigma)$  is  $\rho$ -zCDP.
- ii.  $D_\alpha(\sigma(\cdot|\omega) \parallel \sigma(\cdot|\omega')) \leq \rho\alpha$  for each  $\alpha > 1$  and each  $\omega, \omega' \in \Omega$  with  $1 \leq |\omega - \omega'| \leq T$ .
- iii. We have
$$E_\tau \left[ \left( \frac{\mu(\omega)}{\mu_0(\omega)} \right)^\alpha \left( \frac{\mu(\omega')}{\mu_0(\omega')} \right)^{1-\alpha} \right] \leq e^{\rho\alpha(\alpha-1)} \text{ for each } \alpha > 1 \text{ and } \omega, \omega' \in \Omega \text{ with } 1 \leq |\omega - \omega'| \leq T. \quad (7)$$

Under differential privacy, the distributions of posterior beliefs about the population statistic that the designer can induce are described by two sets of *posteriors*: an  $\epsilon$ -DP mechanism can induce the distribution  $\tau \in \Delta(\Delta(\Omega))$  precisely when  $\tau = \nu \circ P^{-1}$  for some countably supported Bayes-plausible  $\nu$  supported in  $K(\epsilon, \beta_0)$  (Lemma 4), and an oblivious  $\epsilon$ -DP mechanism can induce  $\tau$  precisely when  $\tau$  is countably supported, Bayes-plausible, and supported in  $K_\Omega(\epsilon, \mu_0)$  (Proposition 2). With zCDP, this is no longer true, and we must consider the distributions themselves: for each  $\rho > 0$ , let

$$\begin{aligned} \mathcal{F}(\rho, \beta_0) &\equiv \left\{ \nu \circ P^{-1} \mid \nu \in \Delta(\Delta(\Theta)) \text{ satisfies (6), } \text{supp } \nu \text{ is countable, and } E_\nu \beta = \beta_0 \right\} \text{ and} \\ \mathcal{F}_\Omega(\rho, \beta_0) &\equiv \left\{ \tau \in \Delta(\Delta(\Omega)) \mid \tau \text{ satisfies (7), } \text{supp } \tau \text{ is countable, and } E_\tau \mu = \mu_0 \right\}. \end{aligned}$$

Since a mechanism that induces the distribution of posteriors  $\nu$  induces the distribution  $\nu \circ P^{-1}$  of posterior beliefs about the population statistic, Lemma S.1 implies that  $\mathcal{F}(\rho, \beta_0)$  is the set of such distributions that  $\rho$ -zCDP mechanisms can induce. Likewise, since any countably supported Bayes-plausible  $\tau \in \Delta(\Delta(\Omega))$  can be induced by an oblivious mechanism (Section 4), Proposition S.2 implies that  $\mathcal{F}_\Omega(\rho, \beta_0)$  is the corresponding set for oblivious  $\rho$ -zCDP mechanisms. Since every oblivious mechanism is a data publication mechanism,  $\mathcal{F}_\Omega(\rho, \beta_0) \subseteq \mathcal{F}(\rho, \beta_0)$ ; oblivious mechanisms are without loss under  $\rho$ -zCDP precisely when this inclusion holds with equality. Comparing these sets is thus the zCDP counterpart of comparing  $K_\Omega(\epsilon, \mu_0)$  with  $PK(\epsilon, \beta_0)$  in Theorems 1 and 2.

Theorems S.1 and S.2 show that the conclusions of Theorems 1 and 2 both extend to zCDP: oblivious mechanisms are never without loss of generality with magnitude data, and always without loss of generality with binary data and anonymous respondents.

**Theorem S.1** (Oblivious Mechanisms and Magnitude Data under zCDP). *With magnitude data, for any database prior  $\beta_0$  and any  $\rho > 0$ ,  $\mathcal{F}_\Omega(\rho, \beta_0) \neq \mathcal{F}(\rho, \beta_0)$ . Consequently, there exist  $\rho$ -zCDP data publication mechanisms that induce distributions of beliefs about the population statistic that cannot be replicated by an oblivious mechanism that is also  $\rho$ -zCDP.*

The proof of Theorem S.1 constructs a distribution of posteriors that provides information about the number of respondents in the database with a positive type. With magnitude data, this number can differ across databases with the same population statistic — it is 1 at  $(2, 0, \dots, 0)$  but 2 at  $(1, 1, 0, \dots, 0)$  — while the posteriors that the projected distribution places on each population statistic only provide information about its average across those databases. As in Theorem 1, obliviousness is thus restrictive: of these two databases, only  $(2, 0, \dots, 0)$  is adjacent to the all-zero database  $\mathbf{0}$ , so the posterior can differ more between  $\mathbf{0}$  and  $(1, 1, 0, \dots, 0)$  than it can between  $\mathbf{0}$  and  $(2, 0, \dots, 0)$ , and both enter the average that determines what the decision maker learns about the population statistic.

With binary data, by contrast, averaging a mechanism over permutations of the respondents produces an oblivious mechanism: because databases with the same population statistic are permutations of one another, the average depends on the database only through the population statistic. This averaging preserves the privacy guarantee, and when respondents are anonymous, it leaves unchanged what the decision maker learns about the population statistic.

**Theorem S.2** (Oblivious Mechanisms and Binary Data under zCDP). *With binary data and anonymous respondents,  $\mathcal{F}_\Omega(\rho, \beta_0) = \mathcal{F}(\rho, \beta_0)$  for any  $\rho > 0$ : For each  $\rho$ -zCDP data publication mechanism, there exists a  $\rho$ -zCDP oblivious data publication mechanism that induces the same distribution of posterior beliefs about the population statistic.*

### S.5.1 Omitted Proofs for Section S.5

**Proof of Proposition S.2 ((i) $\Rightarrow$ (ii))** Fix  $\alpha > 1$  and  $\omega, \omega' \in \Omega$  with  $1 \leq |\omega - \omega'| \leq T$ . As shown in the proof of Proposition 2, there exist adjacent databases  $\theta, \theta' \in \Theta$  with  $\omega_\theta = \omega$  and  $\omega_{\theta'} = \omega'$ . Then since  $(S, \sigma)$  is  $\rho$ -zCDP,  $D_\alpha(\sigma(\cdot|\omega) \parallel \sigma(\cdot|\omega')) = D_\alpha(\sigma(\cdot|\omega_\theta) \parallel \sigma(\cdot|\omega_{\theta'})) \leq \rho\alpha$ .

**((ii) $\Rightarrow$ (i))** If  $\theta, \theta' \in \Theta$  are adjacent, then either  $\omega_\theta = \omega_{\theta'}$ , in which case  $\sigma(\cdot|\omega_\theta) = \sigma(\cdot|\omega_{\theta'})$  and  $D_\alpha(\sigma(\cdot|\omega_\theta) \parallel \sigma(\cdot|\omega_{\theta'})) = 0 \leq \rho\alpha$  for each  $\alpha > 1$ , or  $1 \leq |\omega_\theta - \omega_{\theta'}| \leq T$ , in which case (3) follows from (ii).

**((ii) $\Leftrightarrow$ (iii))** Let  $\mu_s \in \Delta(\Omega)$  be the posterior belief about the population statistic induced by output  $s \in S$ , where without loss (as in (4)) distinct outputs induce distinct posteriors. By Bayes' rule,  $\sigma(s|\omega) = \tau(\mu_s)\mu_s(\omega)/\mu_0(\omega)$  for each  $\omega \in \Omega$ ; substituting into (2) as in (5) yields

$$D_\alpha(\sigma(\cdot|\omega) \parallel \sigma(\cdot|\omega')) = \frac{1}{\alpha - 1} \log E_\tau \left[ \left( \frac{\mu(\omega)}{\mu_0(\omega)} \right)^\alpha \left( \frac{\mu(\omega')}{\mu_0(\omega')} \right)^{1-\alpha} \right].$$

Since  $\alpha - 1 > 0$  and  $\log$  is increasing,  $D_\alpha(\sigma(\cdot|\omega) \parallel \sigma(\cdot|\omega')) \leq \rho\alpha$  if and only if (7) holds for  $\omega, \omega'$ , and  $\alpha$ .  $\square$

Theorem S.1's proof uses a distribution of posteriors supported on two beliefs. Given a function  $\psi : \Theta \rightarrow (0, 1)$ , let  $\bar{\psi} \equiv \sum_{\theta \in \Theta} \beta_0(\theta)\psi(\theta)$ , and let  $\nu_\psi$  place probability  $\bar{\psi}$  on the posterior  $\beta^+$  and probability  $1 - \bar{\psi}$  on the posterior  $\beta^-$ , where

$$\beta^+(\theta) \equiv \frac{\beta_0(\theta)\psi(\theta)}{\bar{\psi}} \quad \text{and} \quad \beta^-(\theta) \equiv \frac{\beta_0(\theta)(1 - \psi(\theta))}{1 - \bar{\psi}}.$$

Then  $\nu_\psi$  is Bayes-plausible, since  $\bar{\psi}\beta^+ + (1 - \bar{\psi})\beta^- = \beta_0$ . And since  $\beta^+(\theta)/\beta_0(\theta) = \psi(\theta)/\bar{\psi}$  and  $\beta^-(\theta)/\beta_0(\theta) = (1 - \psi(\theta))/(1 - \bar{\psi})$ , the expectation in (6) is

$$E_{\nu_\psi} \left[ \left( \frac{\beta(\theta)}{\beta_0(\theta)} \right)^\alpha \left( \frac{\beta(\theta')}{\beta_0(\theta')} \right)^{1-\alpha} \right] = \Phi_\alpha(\psi(\theta), \psi(\theta')), \quad (8)$$

where  $\Phi_\alpha(p, p') \equiv p^\alpha(p')^{1-\alpha} + (1-p)^\alpha(1-p')^{1-\alpha}$ . By Lemma S.1,  $\nu_\psi$  can thus be induced by a  $\rho$ -zCDP mechanism if and only if  $\Phi_\alpha(\psi(\theta), \psi(\theta')) \leq e^{\rho\alpha(\alpha-1)}$  for each pair of adjacent databases  $\theta, \theta'$  and each  $\alpha > 1$ . Lemma S.2 provides a pair of probabilities for which  $\log \Phi_\alpha(b, a)/(\alpha(\alpha-1)) \rightarrow \rho$  as  $\alpha \downarrow 1$ : raising the larger of the two violates the zCDP bound.

**Lemma S.2.** *For each  $\rho > 0$ , there exist  $0 < a < b < 1$  such that  $\Phi_\alpha(a, b) = \Phi_\alpha(b, a) \leq e^{\rho\alpha(\alpha-1)}$  for each  $\alpha > 1$ , but for each  $r \in (b, 1)$ ,  $\Phi_\alpha(r, a) > e^{\rho\alpha(\alpha-1)}$  for some  $\alpha > 1$ .*

*Proof.* Since  $\epsilon \tanh(\epsilon/2)$  is continuous and strictly increasing in  $\epsilon$ , from zero to infinity, there is a unique  $\epsilon > 0$  with  $\epsilon \tanh(\epsilon/2) = \rho$ ; let  $a = 1/(1 + e^\epsilon)$  and  $b = 1 - a = e^\epsilon/(1 + e^\epsilon)$ . Since  $b/a = e^\epsilon$  and  $b = 1 - a$ ,

$$\Phi_\alpha(b, a) = be^{(\alpha-1)\epsilon} + ae^{-(\alpha-1)\epsilon} = \frac{e^{(\alpha-\frac{1}{2})\epsilon} + e^{-(\alpha-\frac{1}{2})\epsilon}}{e^{\epsilon/2} + e^{-\epsilon/2}} = \frac{\cosh((2\alpha-1)\epsilon/2)}{\cosh(\epsilon/2)} = \Phi_\alpha(a, b).$$

Hence,  $\log \Phi_1(b, a) = 0$ , and

$$\frac{\partial}{\partial \alpha} \log \Phi_\alpha(b, a) = \epsilon \tanh((2\alpha-1)\epsilon/2).$$

Because  $\tanh$  is concave on  $\mathbb{R}_+$  with  $\tanh(0) = 0$ ,  $\tanh(ux) \leq u \tanh(x)$  for each  $u \geq 1$ ; taking  $u = 2\alpha - 1$ , we have  $\frac{\partial}{\partial \alpha} \log \Phi_\alpha(b, a) \leq (2\alpha - 1)\epsilon \tanh(\epsilon/2) = (2\alpha - 1)\rho$ . Integrating from 1 to  $\alpha$  yields  $\log \Phi_\alpha(b, a) \leq \rho\alpha(\alpha - 1)$ .

For the second claim, note that since  $\Phi_1(p, p') = 1$ , l'Hôpital's rule gives

$$\lim_{\alpha \downarrow 1} \frac{\log \Phi_\alpha(p, p')}{\alpha - 1} = p \log \left( \frac{p}{p'} \right) + (1 - p) \log \left( \frac{1 - p}{1 - p'} \right) = D_{KL}(p \| p')$$

for each  $p, p' \in (0, 1)$ . Since  $b/a = e^\epsilon$  and  $(1 - b)/(1 - a) = e^{-\epsilon}$ , we have  $D_{KL}(b \| a) = \epsilon(2b - 1) = \epsilon \tanh(\epsilon/2) = \rho$ . And since  $\partial D_{KL}(p \| a)/\partial p = \log(p(1 - a)/(a(1 - p))) > 0$  for  $p > a$ , it follows that  $D_{KL}(r \| a) > \rho$  for each  $r \in (b, 1)$ . Hence

$$\lim_{\alpha \downarrow 1} \frac{\log \Phi_\alpha(r, a) - \rho\alpha(\alpha - 1)}{\alpha - 1} = D_{KL}(r \| a) - \rho > 0,$$

and so  $\log \Phi_\alpha(r, a) > \rho\alpha(\alpha - 1)$  for  $\alpha$  close enough to 1.  $\square$

**Proof of Theorem S.1** Fix  $\rho > 0$  and let  $a, b$  be as in Lemma S.2. First observe that whenever  $|\log(p/p')| \leq \rho$  and  $|\log((1 - p)/(1 - p'))| \leq \rho$ , we have

$$\Phi_\alpha(p, p') = p \left( \frac{p}{p'} \right)^{\alpha-1} + (1 - p) \left( \frac{1 - p}{1 - p'} \right)^{\alpha-1} \leq e^{\rho(\alpha-1)} \leq e^{\rho\alpha(\alpha-1)} \quad (9)$$

for each  $\alpha > 1$ . Since  $\log(c/b)$  and  $\log((1-b)/(1-c))$  both vanish as  $c \downarrow b$ , we can choose  $c \in (b, 1)$  with  $\max\{\log(c/b), \log((1-b)/(1-c))\} \leq \rho$  so that (9) holds for both  $(b, c)$  and  $(c, b)$ .

Let  $q(\theta) = |\{i : \theta_i > 0\}|$  denote the number of respondents with a positive type, and let

$$\psi(\theta) = \begin{cases} a, & q(\theta) = 0; \\ b, & q(\theta) = 1; \\ c, & q(\theta) \geq 2. \end{cases}$$

If  $\theta, \theta' \in \Theta$  are adjacent, so that  $\theta_{-i} = \theta'_{-i}$  for some  $i$ , then  $|q(\theta) - q(\theta')| \leq 1$ . Then either  $\psi(\theta) = \psi(\theta')$ , or  $\{\psi(\theta), \psi(\theta')\}$  is  $\{a, b\}$  or  $\{b, c\}$ . In each case,  $\Phi_\alpha(\psi(\theta), \psi(\theta')) \leq e^{\rho\alpha(\alpha-1)}$  for each  $\alpha > 1$ , by Lemma S.2 (when  $\{\psi(\theta), \psi(\theta')\} = \{a, b\}$ ) and (9) (when  $\{\psi(\theta), \psi(\theta')\} = \{b, c\}$ ). By (8), the Bayes-plausible distribution of posteriors  $\nu_\psi$  thus satisfies (6), and so the distribution  $\tau = \nu_\psi \circ P^{-1}$  of posterior beliefs about the population statistic lies in  $\mathcal{F}(\rho, \beta_0)$ .

Now let

$$\psi_\Omega(\omega) \equiv \frac{1}{\mu_0(\omega)} \sum_{\theta: \omega_\theta = \omega} \beta_0(\theta) \psi(\theta)$$

denote the expectation of  $\psi$  conditional on  $\omega_\theta = \omega$ . We have  $P\beta^+(\omega)/\mu_0(\omega) = \psi_\Omega(\omega)/\bar{\psi}$  and  $P\beta^-(\omega)/\mu_0(\omega) = (1 - \psi_\Omega(\omega))/(1 - \bar{\psi})$ . The same computation as in (8) gives

$$E_\tau \left[ \left( \frac{\mu(\omega)}{\mu_0(\omega)} \right)^\alpha \left( \frac{\mu(\omega')}{\mu_0(\omega')} \right)^{1-\alpha} \right] = \Phi_\alpha(\psi_\Omega(\omega), \psi_\Omega(\omega'))$$

for each  $\omega, \omega' \in \Omega$  and each  $\alpha > 1$ . The only database with  $\omega_\theta = 0$  is the all-zero database  $\mathbf{0}$ , so  $\psi_\Omega(0) = a$ . Further, observe that  $\psi(\theta) \in \{b, c\}$  for each  $\theta$  with  $\omega_\theta = 2$ ; then since  $\beta_0$  has full support and  $\omega_{(2,0,\dots,0)} = \omega_{(1,1,0,\dots,0)} = 2$  but  $\psi(2,0,\dots,0) = b$  and  $\psi(1,1,0,\dots,0) = c$ , we must have  $\psi_\Omega(2) \in (b, c)$ . Thus, by Lemma S.2,  $\Phi_\alpha(\psi_\Omega(2), \psi_\Omega(0)) > e^{\rho\alpha(\alpha-1)}$  for some  $\alpha > 1$ . Since  $1 \leq |2 - 0| \leq T$ ,  $\tau$  thus violates (7), and so  $\tau \notin \mathcal{F}_\Omega(\rho, \beta_0)$ : by Proposition S.2, it cannot be induced by an oblivious  $\rho$ -zCDP mechanism. Since  $\mathcal{F}_\Omega(\rho, \beta_0) \subseteq \mathcal{F}(\rho, \beta_0)$ , the inclusion is strict.  $\square$

**Proof of Theorem S.2** Let  $(S, \pi)$  be a  $\rho$ -zCDP mechanism, let  $G$  be the set of permutations of  $\{1, \dots, N\}$ , and for each  $g \in G$ , let  $g\theta \equiv (\theta_{g(1)}, \dots, \theta_{g(N)})$ . Define the mechanism  $(S, \bar{\pi})$  by

$$\bar{\pi}(s|\theta) = \frac{1}{|G|} \sum_{g \in G} \pi(s|g\theta).$$

Fix an ordered pair of adjacent databases  $\theta, \theta'$  and  $\alpha > 1$ . Since  $g\theta$  and  $g\theta'$  are adjacent for each  $g \in G$ , and since  $(x, y) \mapsto x^\alpha y^{1-\alpha}$  is convex on  $\mathbb{R}_+^2$  under the extended-value conventions in (2),

$$\begin{aligned} \sum_{s \in S} \bar{\pi}(s|\theta)^\alpha \bar{\pi}(s|\theta')^{1-\alpha} &\leq \frac{1}{|G|} \sum_{g \in G} \sum_{s \in S} \pi(s|g\theta)^\alpha \pi(s|g\theta')^{1-\alpha} \\ &= \frac{1}{|G|} \sum_{g \in G} \exp((\alpha - 1)D_\alpha(\pi(\cdot|g\theta) \parallel \pi(\cdot|g\theta'))) \leq e^{\rho\alpha(\alpha-1)}, \end{aligned}$$

and so by (2),  $(S, \bar{\pi})$  is  $\rho$ -zCDP. Moreover,  $\bar{\pi}(\cdot|h\theta) = \bar{\pi}(\cdot|\theta)$  for each  $h \in G$ , so  $(S, \bar{\pi})$  is permutation-

invariant; since the data is binary, databases with the same population statistic are permutations of one another (Section 4), and so  $(S, \bar{\pi})$  is oblivious.

Finally, since respondents are anonymous,  $\beta_0(g\theta) = \beta_0(\theta)$  for each  $g \in G$ . And since  $\omega_{g\theta} = \omega_\theta$ , the map  $\theta \mapsto g\theta$  is a bijection from  $\{\theta : \omega_\theta = \omega\}$  to itself. Hence for each  $\omega \in \Omega$  and  $s \in S$ ,

$$\sum_{\theta: \omega_\theta = \omega} \beta_0(\theta) \bar{\pi}(s|\theta) = \frac{1}{|G|} \sum_{g \in G} \sum_{\theta: \omega_\theta = \omega} \beta_0(g\theta) \pi(s|g\theta) = \sum_{\theta: \omega_\theta = \omega} \beta_0(\theta) \pi(s|\theta).$$

Thus  $(S, \pi)$  and  $(S, \bar{\pi})$  induce the same joint distribution of  $\omega$  and  $s$ , and hence the same distribution of posterior beliefs about the population statistic. It follows that  $\mathcal{F}(\rho, \beta_0) \subseteq \mathcal{F}_\Omega(\rho, \beta_0)$ ; the reverse inclusion holds because every oblivious mechanism is a data publication mechanism.  $\square$

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